

Is a Goalkeeper's Lateral or Depth Positioning More Effective at Preventing Goals in Soccer?

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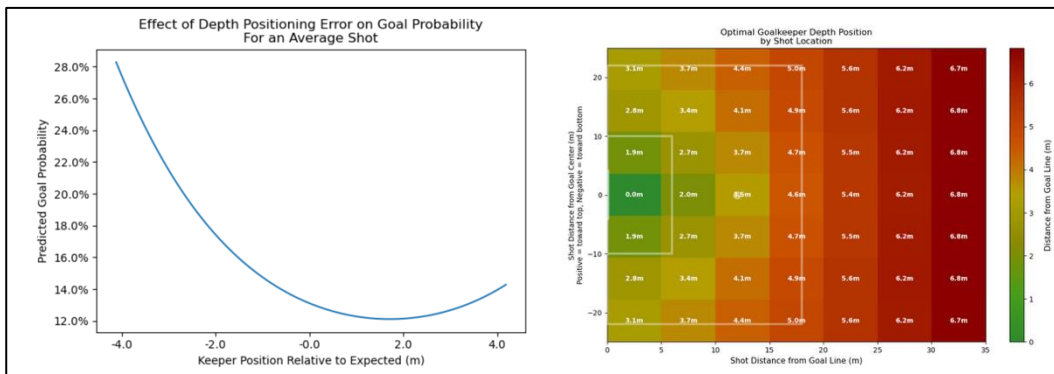
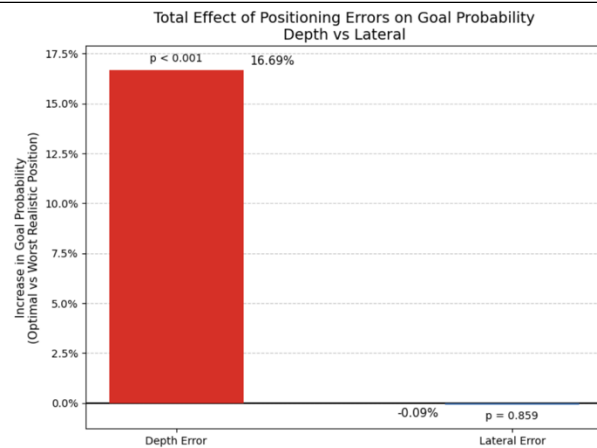
Problem Statement :

When an attacker takes a shot in soccer, the goalkeeper positions themselves to minimize the chance of a goal. In doing so, they must manage two key spatial dimensions: their lateral position and depth from the goal line. However, it is unclear which one is more impactful in preventing goals.

To answer this question, I developed a machine learning model in Python that estimates how goalkeeper lateral and depth positioning affect the probability of a goal. The results from this model can inform goalkeeper coaching by identifying which positional direction goalkeepers should prioritize and their optimal position, potentially reducing the number of goals conceded.

Notes:

- The bar and line charts display probabilities for an average shot, which is taken 18.5 meters from goal with a shooting angle of 26 degrees.
- In the depth positioning chart, positive distance values represent the goalkeeper positioned farther from goal than expected, while negative values represent the goalkeeper positioned closer to goal than expected. Additionally, the minimum value of the graph is the optimal goalkeeper position (1.71 m).



Dataset:

- Data sourced from StatsBomb Open Data GitHub Repo
- Dataset contains event and tracking data for thousands of games (only used event data)

Methodology:

- Retrieved shot data using API calls (~47,000 shots)
- Cleaned and filtered data, adding new variables
- Performed linear regressions between relevant variables and goalkeeper position
- Computed of lateral and depth residuals to isolate goalkeeper positioning from confounding factors
- Fit logistic regression models using the residuals to estimate goal probability
- Generated synthetic data to isolate both residuals
- Plotted goal probability vs residual, created a bar chart to visualize impact of lateral/depth positioning, and made a heatmap to show optimal depth positions for other shots

Results:

- Goalkeeper depth positioning has a larger effect on goal probability than lateral positioning.
- The effect of depth positioning in the logistic regression is statistically significant ($p < 0.001$), while the effect of lateral position is not significant ($p = 0.859$).
- For an average shot, the goalkeeper's depth positioning (within plausible values) can change the goal probability by 16.69%.
- In contrast, lateral positioning can only change goal probability by 0.09%, and the value is statistically insignificant.

Implication:

- The results suggest that the goalkeeper should prioritize being at the optimal depth position over lateral position.

Sources

Sites Used: Jupyter Notebook (Coding), GitHub (to access StatsBomb Open Data), Claude (helped overall process)